MIL-S-52302C
13 October 1983
SUPERSEDING
MIL-S-52302B
18 February 1970

MILITARY SPECIFICATION

STRUT, BRACING, TRESTLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a trestle bracing strut to be used with the trestle assembly (MIL-T-52150).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel and Seals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coating: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13207E9733	- Strut, Bracing, Trestle.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The strut shall be as shown on Top Assembly TA13207E9733 and as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the strut. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be new and unused.

3.3.1 Material deterioration and control. The strut shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-C-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of Materials and Finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, shall make information available, upon request, to the contracting officer or designated representative.

3.4 Identification marking. The strut shall be metal die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.5 Treatment and painting. The inside and outside surfaces of the strut shall be cleaned, treated, and painted in accordance with MIL-T-704, type B.

3.6 Stenciling. The nomenclature, National stock number, weight, and cube shall be stenciled on each strut. The letters "NET WT" shall precede the numerals in the weight and letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

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3.7 Workmanship. Holes shall be drilled through both walls of the tubing in one operation through a metal template, jig, or fixture so as to be alined true, square, and normal to the tubing. All burrs, rough edges, and edges of holes shall be ground smooth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disaproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article strut shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure. The strut shall be examined as specified herein for the following defects:

4.5.1 Examination.

Major

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and/or operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Workmanship not as specified.
106. Any part not in accordance with the QAP requirements as shown on the drawings.

Minor

201. Dimensions other than those identified by QAP data not as specified.
202. Identification marking incorrect, missing, or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.

4.6 Inspection of packaging. The packing and marking shall be examined to determine compliance with section 5 of this specification. Strapping for level A that is not in accordance with the referenced document or any marking for level A or commercial that is missing, illegible, incorrect, or incomplete shall be cause for rejection.

5. PACKAGING

5.1 Packing (see 6.6). Packing shall be level A or commercial as specified (see 6.2).

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5.1.1 Level A. The struts shall be bundled together in quantities of six. Each bundle shall be secured with not less than five flat steel straps conforming to QQ-S-781, type I or IV, finish B, size 3/4 by 0.020. The straps shall be placed approximately 18 inches from each end of the bundle, and the intermediate straps shall be equally spaced between the end straps.

5.1.2 Commercial. Struts shall be packed in accordance with MIL-STD-1188.

5.2 Marking (see 6.6).

5.2.1 Military. Marking for level A packaging shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The strut is intended for use in floating and fixed bridges for bracing trestles to obtain lateral and longitudinal stability.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. Degree of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item shall be examined and should be a preproduction strut. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the documents first article.

6.4.1 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

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6.5 Definition.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Packing. Degrees of preservation have not been included as the struts do not require additional preservation. For purposes of preservation/packing level marking, the preservation shall be designated as level A.

Custodian:
Army - ME

Review activity:
Navy - MC

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-52302C	Strut, Bracing, Trestle

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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